Tectonic History of the Ethiopian Rift as Deduced by K-Ar Ages and Paleomagnetic Measurements of Basaltic Dikes

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The tectonic history within a continental rift is inferred from K-Ar ages and paleomagnetic measurements from oriented samples of basaltic dikes. The stresses required to produce the strain distribution (dikes) are consistent with extension in the plane of the earth, which varies episodically as a function of geologic time; furthermore, an average paleomagnetic pole is calculated to be consistent with the present-day pole position and indicates that large-scale rotation of African continental blocks is untenable. An average rate of continental crust of ~2 cm/10^4 years is generated over a 70-km traverse along the western margin of the Ethiopian rift.

In view of the recent discussions of plate tectonics and sea-floor spreading as evidenced by magnetic anomalies and seismic and heat-flow data from the ridges of the ocean floor, it seems worthwhile to examine analogous features in a continental environment, where geologic sampling and controls are more easily obtainable. By understanding the tectonic history from a continental-rift environment, we should expect to gain additional insight into the dynamics of the earth's crust.

In the present study we assume that tectonics, or fracturing of the earth's crust, and igneous activity are intimately related. Therefore by measuring the distribution in time and space of magma intrusion into fractures of the earth's crust to form dikes we can infer the stress patterns and the relative motions of large crustal masses and thereby deduce the tectonic history of the region, provided, of course, that subsequent crustal movement after dike formation can be recorded. For assistance in answering this particular question we use paleomagnetic data to record the variation in the distribution of the magnetic dipoles of the dikes as a function of geologic time.

Additional and important information about the petrological and geochemical distribution of the dikes as functions of time and space will be treated more extensively in a future publication.

FIELD DATA

The area chosen for this study is in Ethiopia (Figure 1), where the Ethiopian rift valley divides the region into a western and an eastern plateau. South of Ethiopia the rift valley extends into Kenya and Tanzania, and to the north the rift valley widens where it intersects the Red Sea and the Gulf of Aden. This northern region is called Afar, the geology of which has recently been reviewed [*Mohr*, 1970].

Generally the plateau regions of Ethiopia consist of alkali basalts that are primarily flood basalts belonging to the trap series. These flood basalts are believed to be Eocene-Oligocene in age. Of great importance tectonically is the fact that these series of lavas are downwarped and

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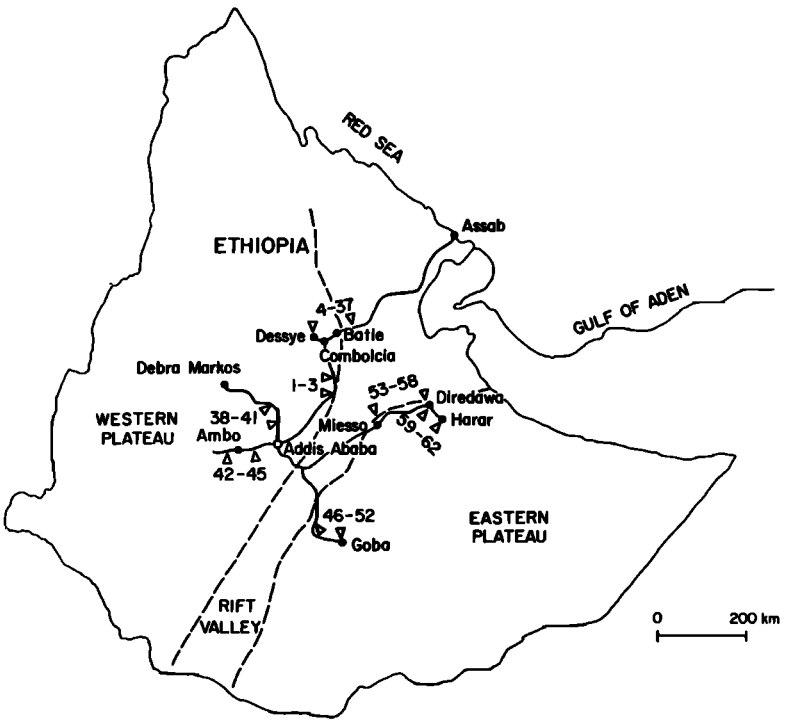


Fig. 1. Index map of oriented dike samples collected in Ethiopia.

thicken at the rift margins. A volcanological sketch map of the region reviewing the estimated ages of volcanism in the Ethiopian rift and plateaus has been published [*Mohr*, 1971]. Generally the youngest volcanics are found in the Afar region and along the central axis of the rift valley, the older volcanics being exposed in the plateau areas.

Because of the rapid development of lateritic soil in this region, fresh rock is exposed almost exclusively along relatively recent road excavations. Consequently, the most efficient manner of locating and collecting dike samples is to practice armchair geology from a Land Rover. Therefore all the sampled dike localities are situated either along or within 0.5 km of the major road system (Figure 1).

From a total of almost 200 sampled dike localities in Ethiopia, 62 oriented samples were collected from the regions shown in Figure 1. Samples 1–45 were collected from the western plateau of the rift system, and samples 46–62 were collected from the eastern plateau region.

On the western plateau, samples were collected along a traverse across the rift margin from Dessye (elevation of 2500 meters) going eastward down the escarpment to Combolcia (1900 meters), then to Batie (1680 meters), and finally to the rift floor at an elevation of 650 meters. Additional oriented dike samples were collected from the interior of the western plateau along the Debra Markos and Ambo roads, which are northwest and west, respectively, of Addis Ababa. Rocks from these areas are stratigraphically equivalent to those from the region west of Dessye.

From the eastern plateau region, oriented samples were collected from basaltic dikes on a traverse across the margin of the rift going from the floor of the rift valley at Diredawa to the increased elevation of Harar. Here the dikes traverse crystalline basement rocks. Between Miesso and Harar oriented dike samples were collected along the rim of the eastern plateau. In the Goba region, basaltic dikes were sampled in the Batu mountains just west of Goba. No dikes were encountered along the margin of the eastern plateau on the Goba road before the Batu mountains.

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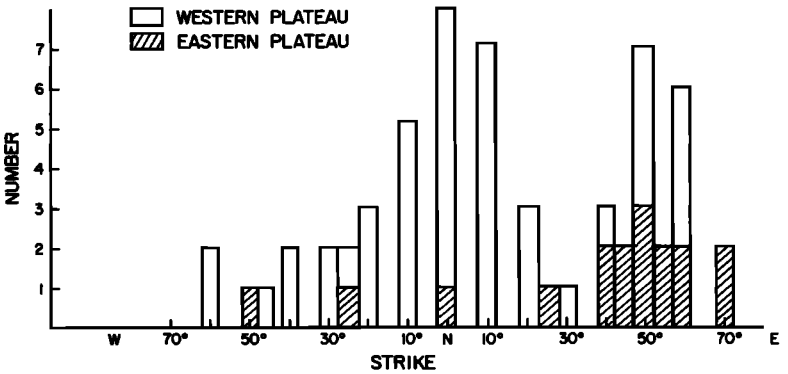


Fig. 2. Strike orientation of Ethiopian dikes.

The strikes of the basaltic dikes, from which oriented samples were collected on both the western and the eastern plateau of Ethiopia, are summarized in Figure 2. On the western plateau the distribution of the strike orientations of the dikes is concordant with the main tectonic features of the area. The predominant orientations are consistent with the rift valley trend of N-S, N10°E, and N10°W. Furthermore, the distribution of the strike orientations has maximums at N50°E and possibly at N30°–50°W, which are similar to the trends of the Gulf of Aden and the Red Sea, respectively.

Within the eastern plateau the dikes are oriented predominantly N50°E, this orientation likewise being consistent with the trend of the Gulf of Aden; however, a few sporadic cases are oriented N-S, N25°W, and N50°W.

The dips of the sampled dikes are nearly always vertical (±10°), and in those cases where the dikes dip significantly from vertical, the dip orientation remains nearly normal (±10°) to the bedding of the host rock.

Experimental Procedure

Before the samples from the dike outcrops in the field were removed each sample was oriented by marking a horizontal plane on the sample. This orientation was determined by a bubble level mounted on a brass plate. In conjunction with the horizontal markings a reference direction as determined by a Brunton compass was also noted on each sample.

In the laboratory the samples were cored normal to the horizontal plane, and in a few cases additional samples were cored parallel to the horizontal reference surface. The remnant magnetic poles were determined after field demagnetization was alternated at 150 oe.

Samples were prepared for potassium and argon analyses by crushing between 5 and 10 grams of the cored rock. Splits were made of the crushed sample. The samples used for potassium determination were ground to a fine powder in a steel mortar with a Spex mixer mill. Duplicate samples weighing 50–100 mg were dissolved in hydrofluoric acid and sulfuric acid, and the excess acid was removed by evaporation. The remaining salts were dissolved in dilute nitric acid and diluted to a 10-ml volume. The potassium was determined by X-ray fluorescence by using a standard addition technique. The potassium $K\alpha$ X ray was counted, a weighed amount of potassium nitrate was added, and the solution was recounted. Duplicate measurements gave a precision of the order of 2%.

Radiogenic argon was determined by melting samples between 0.25 and 0.5 gram in an ultra-high vacuum system. The samples were melted by RF induction heating, and the extracted gases were gettered during melting by allowing the gases to react with hot titanium. Calibration of the system was done by standard addition; i.e., known aliquots of atmospheric argon were added to the extraction system and the mass spectrometer. Duplicate analyses were made on all the samples. The degree of reproducibility of the measurements was primarily a result of the atmospheric argon correction that had to be applied to the data.

Thin sections were prepared from all the collected rocks, and, although the petrological description of the dike rocks will be the subject

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of a future publication, the primary concern at present is the degree of alteration to which the rocks were subjected. In particular, zeolitization, deuteric alteration, and weathering certainly can influence the crystallization ages of the dikes (if they are determined by the K-Ar method) as well as the chemical classification of the rocks (if it is based on abundances of the alkali elements).

RESULTS

K-Ar ages. The measurements determined on the 62 oriented samples collected from various Ethiopian basaltic dikes are summarized in Table 1. Also included with these data are the K-Ar age determinations of seven basalt flows that are hosts to seven of the sampled dikes. The host rock ages were measured as a preliminary check on the age determinations of

TABLE 1. K-Ar Ages of Basaltic Dikes from Ethiopia

Sample No.	Strike	Dip	K, wt.%	$^{40}Ar_{rad}$, $10^{-8}cm^3/g$ STP	$^{40}Ar_{rad}$, %	Age, m.y.
1	N10°E	90°				
2	N10°E	90°				
3	N20°W	80°W				
4	N3°E	70°W	1.48	107 ± 20	65	18.5
5	N-S	90°	0.91	85 ± 7	70	23.5
5*			0.79	76 ± 6	25	24.2
6	N60°E	60°S				
7	N60°E	75°N				
8	N8°E	90°	1.53	138 ± 5	30	22.6
9	N20°E	80°W	2.02	550 ± 50	40	66
10	N50°E	90°	0.68	55 ± 2	50	20.5
11	N60°E	90°	1.20	100 ± 2	40	21
11*			1.56	158 ± 2	35	25.5
12	N10°W	90°	0.67	58 ± 3	30	21.6
13	N15°W	90°	0.81	83 ± 3	60	26
14	N30°W	90°	0.68	30 ± 3	45	11.1
15	N40°W	80°S	1.18	50 ± 5	36	10.8
16	N45°W	65°S	0.86	46 ± 6	40	13.5
17	N50°E	90°	0.34	66 ± 8	50	48.5
18	N20°E	75°W	0.42	58 ± 2	55	35
19	N20°W	55°S				
20	N40°W	90°	1.08	154 ± 8	60	36
21	N60°W	90°	0.72	44 ± 3	40	15.5
22	N30°E	60°W	1.02	127 ± 5	24	32
23	N10°E	60°W	1.28	107 ± 3	40	21.2
24	N10°W	65°W	0.29	34	40	30
25	N15°W	75°W				
26	N10°W	70°W	1.03	230 ± 10	45	55
27	N30°W	80°E	1.72	165	50	24.5
28	N10°W	45°W				
29	N55°E	85°W	1.21	120 ± 3	75	25
29*			1.66	187 ± 2	35	28
30	N40°E	85°W	0.79	77 ± 3	30	25
31	N-S	85°W				
31*			0.37	85 ± 5	50	57
32	N-S	80°W	1.65	122 ± 10	50	19
33	N10°E	70°W	0.73	56 ± 3	40	19.4
34	N50°E	80°E	0.20	89 ± 6	20	(?)
35	N60°W	90°	1.15	116 ± 8	16	25
36	N20°W	85°W	0.66	81 ± 3	32	31
37	N10°W	90°	0.84	10.7 ± 2	15	3.2
38	N60°E	90°				

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TABLE 1. (continued)

Sample No.	Strike	Dip	K, wt.%	$^{40}Ar_{rad}$, $10^{-8}cm^3/g$ STP	$^{40}Ar_{rad}$, %	Age, m.y.
39	N-S	90°	0.97	126 ± 10	65	32
39*			0.93	154 ± 10	40	42
40	N10°E	90°	1.60	123 ± 7	70	19.3
41	N-S	90°	1.02	126 ± 8	50	31.5
42	N-S	90°	0.432	99 ± 4	30	58
43	N20°E	45°E	1.73			
44	N10°E	65°E	0.99	87 ± 5	65	22.5
45	N25°W	75°N	2.66	335 ± 15	60	31
46	N60°E	90°	4.42	330 ± 15	70	19
47	N70°E	60°W	1.22	72 ± 5	50	15
48	N45°E	90°	1.71	100 ± 10	70	15
49	N60°E	75°N	1.41	88 ± 2	60	16
50	N50°E	90°	1.84	140 ± 8	80	19
50*			1.22	145 ± 15	55	30
51	N50°E	90°	1.98	165 ± 8	80	21
52	N55°E	90°	1.87	115	52	15.5
53	N45°E	90°	1.98	87 ± 8	60	11
54	N50°E	90°	1.29	75 ± 5	50	15
54*			0.27	38 ± 3	45	35.5
55	N55°E	90°				
56	N25°W	80°S	0.336			
57	N25°E	70°S	0.235	15.5	1	18
58	N50°W	90°	0.72	79 ± 3	6	28
59	N40°E	85°W	0.52	73 ± 5	50	35
60	N40°E	90°	0.30			
61	N-S	90°	0.81	97 ± 8	13	30
62	N70°E	85°S	0.31	35 ± 3	10	28

* Host rock.

the dikes, since we can expect that the host rocks should be older than the dikes traversing them.

The percentage of radiogenic argon from the basalts generally varies between 30 and 60%, although from a few samples the percentage of radiogenic argon was as low as 10–25%. The absolute abundance of the radiogenic argon varied, being on the average, 5%, although again in a few cases this variation was as low as 2% and as high as 10%. The K-Ar ages, which were calculated by using a $\lambda_e = 0.585 \times 10^{-10}$/year and a branching ratio of 0.121, are within 1% of the K-Ar ages calculated from the nuclear constants as summarized by *Beckinsale and Gale* [1969]. The uncertainty of the K-Ar ages introduced by the variation in the potassium and radiogenic ^{40}Ar abundances is greater than the variation in age introduced by the uncertainty in the nuclear constants.

A plot of the K-Ar age versus the potassium content of the basaltic dikes (Figure 3) displays a scatter in crystallization ages from 65 to 3 m.y. More importantly, the potassium content of the rocks can be seen to range over an order of magnitude for rocks of the same crystallization age. Furthermore, crystallization ages of the seven host rocks (designated *a*) are greater than those of the dikes that traverse them. These observations, along with the fact that the oldest K-Ar ages do not always coincide with rocks of low potassium abundance, lead us to infer that inherent radiogenic argon, or the leaching of potassium from these basalts subsequent to their time of crystallization, does not appear to have altered their 'whole-rock' ages. However, this problem will be discussed more fully in context with the geological localities and the interpretations of the dike ages.

Western plateau. The K-Ar ages of basaltic

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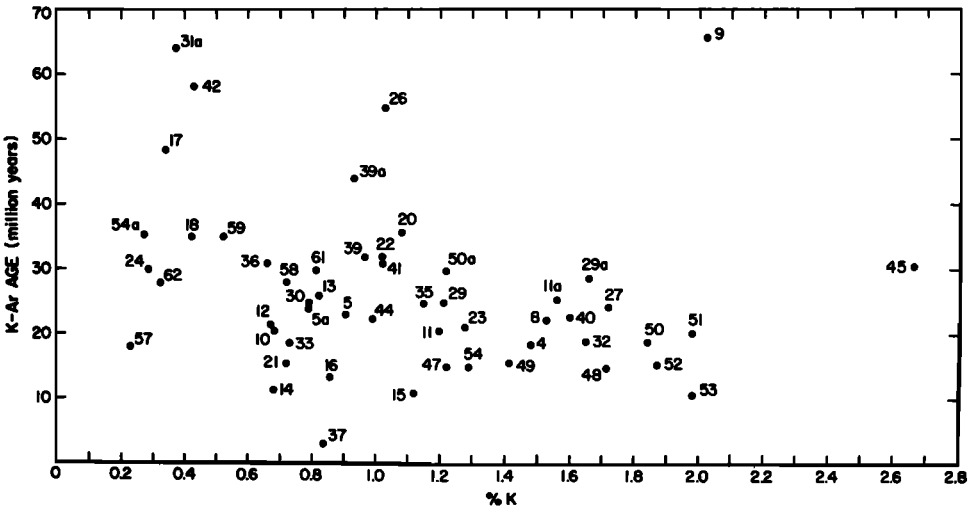


Fig. 3. K-Ar ages versus weight per cent potassium from Ethiopian dike rocks.

dikes sampled along a traverse going from west to east down the margin of the rift valley from Dessye, Combolcia, and Batie and finally to the rift valley floor are shown schematically in cross section (Figure 4). The locations and the dips of the dikes have been projected from their locations as mapped on aerial photographs. Two dikes within 5 km of Dessye have K-Ar ages of 22.6 and 23.5 m.y. A K-Ar age of 24.2 m.y. was measured on the host rock from which the 23.5-m.y. dike was sampled. One of the oldest dikes was found near Combolcia and has a K-Ar age of 66 m.y. From the differences in the K-Ar ages of the dikes at Dessye and Combolcia as well as their respective differences in elevation we estimate the average rate of igneous deposition for the plateau basalts in this area to be approximately 15 m/m.y.

Approximately 10 km east of Combolcia four dikes within 5 km of one another have K-Ar ages of 20.5, 21, 21.6, and 26 m.y. The host rock of the 21-m.y. dike has a K-Ar age of 25.5 m.y.; consequently, the relative stratigraphic position and the age of this host rock in relation to the age of the 66-m.y. dikes at Combolcia strongly indicate that major faulting (downthrown east) has occurred between these two outcrops. This fault is schematically shown on the cross section with the relative motions as deduced from the crystallization ages of the dikes.

Approximately 5 km west of Batie the distribution in the K-Ar ages of the dikes varies between 11 and 48 m.y. Because three dikes within this dike swarm have K-Ar ages of <25 m.y., which is the age of the host rock approximately 20 km west of Batie, a major fault zone (upthrown east) has been inferred and is shown only schematically in cross section as

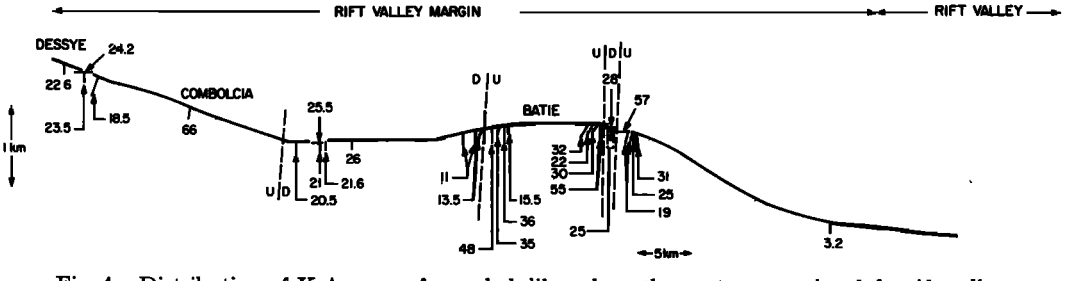


Fig. 4. Distribution of K-Ar ages of sampled dikes along the western margin of the rift valley.

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deduced from the ages of the dikes. The fault zone could well be located 5 km west of where it is actually shown.

Approximately 7 km east of Batie the age distribution of 10 exposed dikes varies between 19 and 55 m.y. The host rocks from two localities east of Batie have K-Ar ages of 28 and 57 m.y. Consequently, the relative stratigraphic positions of the 28-m.y. host flood basalt, which is situated 1 km from a 55-m.y. dike and another kilometer from a 57-m.y. host flood basalt, can be understood if a graben structure with vertical displacements of up to 600 meters exists within the area. This structure has been shown schematically in the cross section (Figure 4).

The youngest basaltic dike (3.2 m.y.) is situated approximately 28 km east of Batie in the floor of the rift valley.

The relative spatial and temporal distributions of the dikes and the flood basalts along the Dessye-Combolcia-Batie traverse indicate that (1) tectonically, the region can be understood as a graben-horst structure, which implies that the principal stress directions that produced this strain pattern originated by an extension in an approximate E-W direction, essentially tangential to the earth's surface (the directions and the episodic nature of this extension will be discussed more fully later), (2) the spatial distribution of the dikes as a function of crystallization age is not a simple relationship progressing from oldest to youngest in a westerly to easterly direction, and (3) the chemical compositions of the dikes as deduced by a variation in potassium abundances are not simply related to either their spatial distribution as measured along the margin of the rift valley or their distribution in crystallization ages.

From the interior of the western plateau four oriented dike samples were collected along the Debra Markos road (Figure 1). The K-Ar ages of samples 39–41 are 32, 19.3, and 31.5 m.y., respectively. In thin section all these samples appear unaltered. Samples 39 and 41, which are petrographically similar, give concordant K-Ar ages; moreover, the host rock from which sample 39 was collected has a K-Ar age of 42 m.y. Because dike sample 40 can be distinguished petrographically from samples 39 and 41, its K-Ar age of 19.3 m.y. is believed to be significantly different from that of the 32-m.y. dikes.

The K-Ar ages of the dikes and the host rocks along the Debra Markos road are 10 and 20 m.y. greater, respectively, than those of the dikes and the host rocks of the Dessye region. These differing ages imply that the stratigraphic horizons are not equivalent for the two regions and that this condition may be the result of faulting or a graben-horst structure similar to the structure along the rift valley margin.

From the Ambo region (Figure 1) oriented samples collected from dikes 42–45 are all coarse-grained augite basalts with augite crystals 3–4 mm long and varying amounts of plagioclase in the groundmass of the rock. The K-Ar ages range between a high of 58 m.y. and a low of 22 m.y. with one intermediate value of 31 m.y. These K-Ar ages are consistent with those determined for the other dikes along the Debra Markos road and also along the rift margin between Dessye and Batie. The K-Ar age of 58 m.y. for one of the dike samples is consistent with a K-Ar age determined for a flood basalt (31*) just east of Batie. Moreover, both samples contain similar potassium abundances of 0.43 and 0.37%, respectively. Consequently, if these K-Ar ages are real crystallization ages and not the result of systematic errors caused by contamination from inherent radiogenic argon, we must again infer that faulting and a graben-horst structure exist within the Ambo region.

Eastern plateau. Approximately 37 km east of Adaba, which is situated on the road to Goba (Figure 1), seven oriented dike samples and one host basalt were collected from the Batu mountain area. Petrographically, these dike rocks contain coarse-grained feldspar laths that frequently are aligned in a flow structure. Minor amounts of rock alteration include sericitization, chloritization, and zeolitization. The potassium abundances of the rocks range from 1.2 to 2.0%. Four of the dikes have K-Ar ages of 15–16 m.y., whereas two other dikes have K-Ar ages of 19 and 21 m.y. The host basalt through which dike 50 traverses has a K-Ar age of 30 m.y.

Farther north, along the rim of the eastern plateau, oriented dike samples 53–58 were collected between Miesso, Kolubbi, and Harar (Figure 1). Dike samples 53 and 54 (Table 1)

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contain 2.0 and 1.3% potassium, respectively, whereas the other four dike samples and one host basalt contain 0.2–0.7% potassium. The K-Ar ages of dikes 53 and 54 are 11 and 15 m.y., respectively. Moreover, the K-Ar age of the host basalt through which the 15-m.y. dike traverses is 35.5 m.y. These crystallization ages of dike and host rock are similar to the relative age differences, which were measured for the host rock and dikes from the eastern plateau region in the Batu mountains. Dike samples 56 and 57, which have 0.24 and 0.34% potassium, respectively, contain a large percentage of atmospheric argon, and consequently a very reliable determination of the radiogenic argon content is prohibited. Sample 58 appears unaltered in thin section and has a K-Ar age of 28 m.y.

As the eastern margin of the rift valley from the rift valley floor at Diredawa was ascended, four oriented samples (59–62) were collected from basaltic dikes traversing the crystalline basement rocks. The oldest dike (35 m.y.) is situated at the lowest elevation near Diredawa, and the younger basalt dikes occur at a higher elevation as the rift valley margin is ascended. Field relationships provide a relative check on the crystallization ages of dikes 61 and 62, since we can expect that dike 62, which was found to traverse dike 61, should have a younger crystallization age. Dike 60, unfortunately, contains a large percentage of atmospheric gas (similar to dike 56), and consequently a determination of the radiogenic argon content was prohibited.

In summary, the distribution of the crystallization ages of the dikes from the eastern margin, the rim, and the plateau surface varies from 11 to 35 m.y. The youngest dikes are not confined to the margin of the rift valley but are found on the plateau surface some 20 km east of the rift margin. The variation in the potassium abundances of the dikes suggests that two distinct petrographic provinces exist in the eastern plateau region. These regions can be conveniently labeled by $K < 0.8\%$ and $K > 1.25\%$. Although there does not appear to be an age difference between these two petrographic rock types, there is a strong suggestion that they may be related to a distribution in space that is primarily a function of the relative distance from the rift valley margin. Further investigation, however, is needed before a firm conclusion can be reached.

PALEOMAGNETIC MEASUREMENTS

The remnant magnetism of the oriented samples was measured and subjected to alternating field demagnetization in successively increasing fields of 50, 100, 150, 200, and 400 oe. As is typical of dike samples containing essentially unoxidized opaque grains of titanomagnetite, the remnant magnetism is quite unstable and decreases rapidly in alternating fields. These samples are unstable enough that no useful paleomagnetic data could be obtained from many of them. The orientation of only 26 samples could be reliably determined. The criterion used for acceptance of the measurements was that the magnetic orientation did not change direction significantly in alternating fields of 100 and 150 oe. Higher fields generally introduced the scattering typical of demagnetizing very soft samples.

The K-Ar ages and the stable magnetic orientations of 19 dike samples from the western plateau are summarized in Figure 5. Six of the samples have reversed polarity, and two have similar K-Ar ages of 19 and 19.4 m.y. From the eastern plateau only seven samples contain stable remnant magnetism (Figure 6), and three of these have reversed polarity.

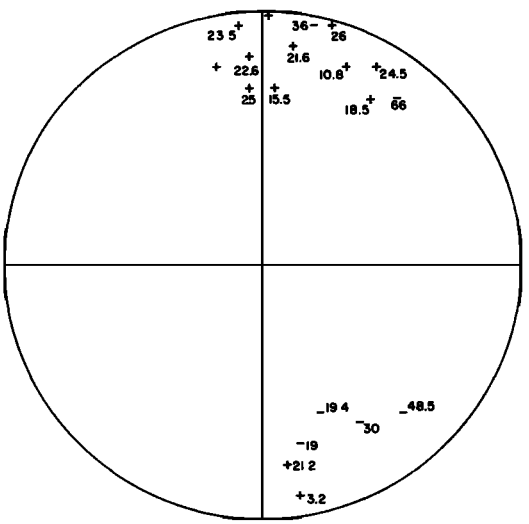


Fig. 5. Paleomagnetic orientations and K-Ar ages from dikes on the western plateau.

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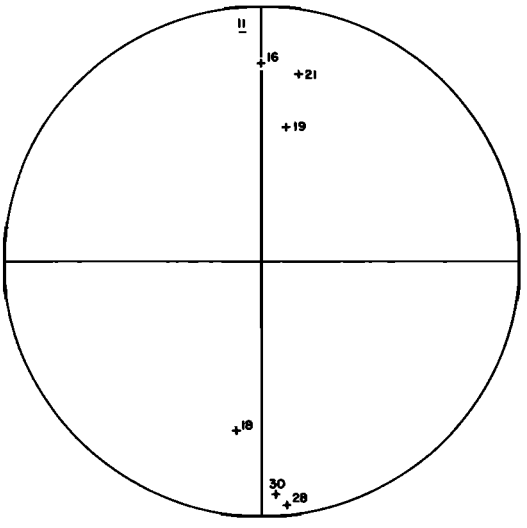


Fig. 6. Paleomagnetic orientations and K-Ar ages from dikes on the eastern plateau.

Because the number of magnetically oriented samples of equivalent K-Ar ages is so small, it is not possible to determine the reversal sequence as a function of geologic time. However, an averaged pole position from all dike samples of the eastern and western Ethiopian plateau, regardless of differences in their crystallization ages, reveals that pole positions for both sides of the rift valley are similar (Table 2) and, furthermore, that this average pole position is consistent with the earth's present magnetic pole. These results give pole positions similar to those determined previously by workers studying lava flows from this part of the world [*Irving and Tarling*, 1961; *Brock*, 1968; *Tarling et al.*, 1967]. These facts strongly suggest that tectonic movement of the dikes subsequent to their formation has been minor and that large-scale shear movement of either the western or the eastern plateau has been negligible, since any such movement would tend to rotate the magnetic poles away from their original position. For example, if the movement along the rift valley was predominantly left sinistral shear such that the eastern margin of the rift valley was rotating in a clockwise direction, we might expect the magnetic dipoles of the dikes from the eastern plateau region to be systematically shifted eastward.

DISCUSSION

A histogram of the crystallization ages of the dikes from the eastern and western plateau regions of Ethiopia (Figure 7) reveals that formation of the dikes is essentially episodic with a periodicity of between 2 and 5 m.y. over the time interval 11–35 m.y., as is evidenced by peaks at 11, 15, 19–21, 23, 25, 27, 31, and 35 m.y. Furthermore, the similarity of the age distribution of the dikes from the eastern and western plateaus indicates that formation of the dikes took place contemporaneously within both regions.

The distribution of the strike orientations of the dikes as a function of geologic age (Figure 8) displays a pattern analogous to that of fractures produced in a rock system that has undergone shear fracture [*Megrue et al.*, 1971]. Consequently, if we assume that formation of the dikes is directly related to crustal rupture and, furthermore, if we employ the Coulomb-

TABLE 2. Summary of Paleomagnetic Measurements from the Western and Eastern Plateau Dike Rocks of Ethiopia

	No. of Samples	D	I	κ	α_{95}	Pole Position Lat.	Long.	δM	δP
Western plateau	19	1	13	9.7	11.4	161°	86°	12	6
Eastern plateau	7	2	5	9.2	21.1	155°	82°	21	11

D, declination of mean direction of remanence; I, inclination of mean direction of remanence; κ, Fisher's precision parameter; α_{95}, radius of circle of confidence containing 95% of individual measurements.

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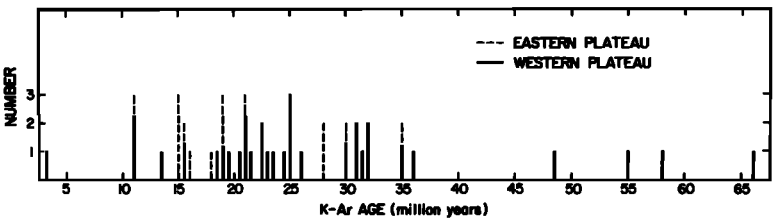


Fig. 7. Distribution of K-Ar dike ages from the eastern and western Ethiopian plateaus. The increased frequency of ages at discrete time intervals suggests that dike emplacement is episodic.

Mohr failure criterion (more recently reviewed by *Handin* [1969]), we can deduce the principal stress directions required to produce the observed strain relationships (dikes), if it is assumed that the effect of the intermediate stress direction σ_2 is negligible. The orientation of the dikes of >30 m.y. is consistent with a minimum principal stress direction σ_3 (extension), oriented essentially in an E-W direction, and a maximum principal stress direction σ_1, oriented in a N-S direction. Both σ_1 and σ_3 lie within the horizontal surface of the earth.

The orientations of the two 28-m.y. dikes are consistent with the trend of the Red Sea and the Gulf of Aden. Both dikes are situated on the eastern plateau between Diredawa and Harar, this location possibly indicating that they resulted from extension in a NNE-SSW direction.

The relative orientations of the 24- to 26-m.y. dikes were measured within the Dessye-Batie traverse and do not lend themselves to a unique interpretation of the stress orientations. One acceptable interpretation is that σ_1 is oriented in a NNE-SSW direction.

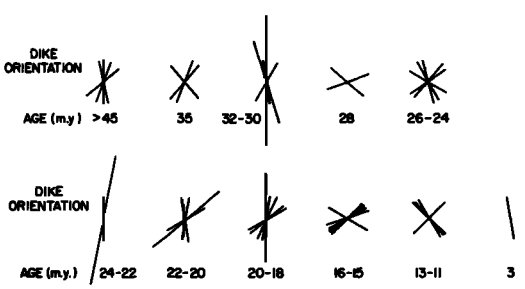


Fig. 8. Distribution of the strike orientations of Ethiopian dikes as a function of crystallization ages. Changes in strike are attributed to changes in the maximum and minimum principal stress directions.

The orientations of the dikes in the three age groupings extending from 24 to 18 m.y. and measured for both the eastern and the western plateau region can be interpreted as resulting from extension in a NNW-SSE direction. The 15- to 16-m.y. dikes, which cluster N50°E, are all from the eastern plateau region, whereas the one dike oriented N50°W is from the western plateau. This strain pattern could result from extension in a N-S direction. Likewise, the strain pattern indicated by the orientation of the 11- to 13-m.y. dikes is consistent with the Red Sea trend from four dike localities on the western plateau. The one dike normal to this trend is from the eastern plateau region.

From the foregoing discussion it seems reasonable to conclude that the history of the tectonic activity within the Ethiopian rift can be explained as resulting from σ_1 and σ_3, which lie within the plane of the earth and change direction as a function of geologic time. This interpretation could explain the anomalous positions of the Danakil and Aisha horsts in the Afar region.

An estimate of the average rate of crustal extension over a time interval of 50 m.y. can be obtained by considering the following observations. The distance along the Dessye-Combolcia-Batie traverse to the floor of the rift valley is approximately 80 km by air. On this traverse 76 dikes with an average thickness of 1.5 meters were sampled. Consequently, we calculate for this traverse a minimum rate of crustal generation of 2.3 m/10^6 years.

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